

# Homework 3

EE 999 — Introduction to Control Systems

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*Due Wednesday, September 23, 2026*

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## Problem 1 — Matlab

Code: ./hw\_03\_01.m

```
1 if exist('OCTAVE_VERSION', 'builtin')
2     pkg load control;
3 end
4
5 % Note these values were chosen to accentuate the differences
6 Kp = 1000;
7 Ki = 1000;
8 Kd = 1000;
9
10 %=====
11
12 m = 1000;
13 b = 50;
14 r = 1;
15
16 s = tf('s');
17 P_cruise = 1/(m*s + b);
18 t = 0:0.1:20;
19
20 %=====
21
22 C = pid(Kp);
23 T_P = feedback(C*P_cruise, 1);
24
25 %=====
26
27 C = pid(Kp,Ki);
28
29 T_PI = feedback(C*P_cruise, 1);
30
31 %=====
32
33 C = pid(Kp, Ki, Kd);
34
35 T_PID = feedback(C*P_cruise, 1);
36
37 %=====
38
39 figure
40 step(r*T_P, 'b', r*T_PI, 'r', r*T_PID, 'g', t)
41 axis([0 20 0 1.5*r])
42 legend('P', 'PI', 'PID', 'Location', 'southeast')
```

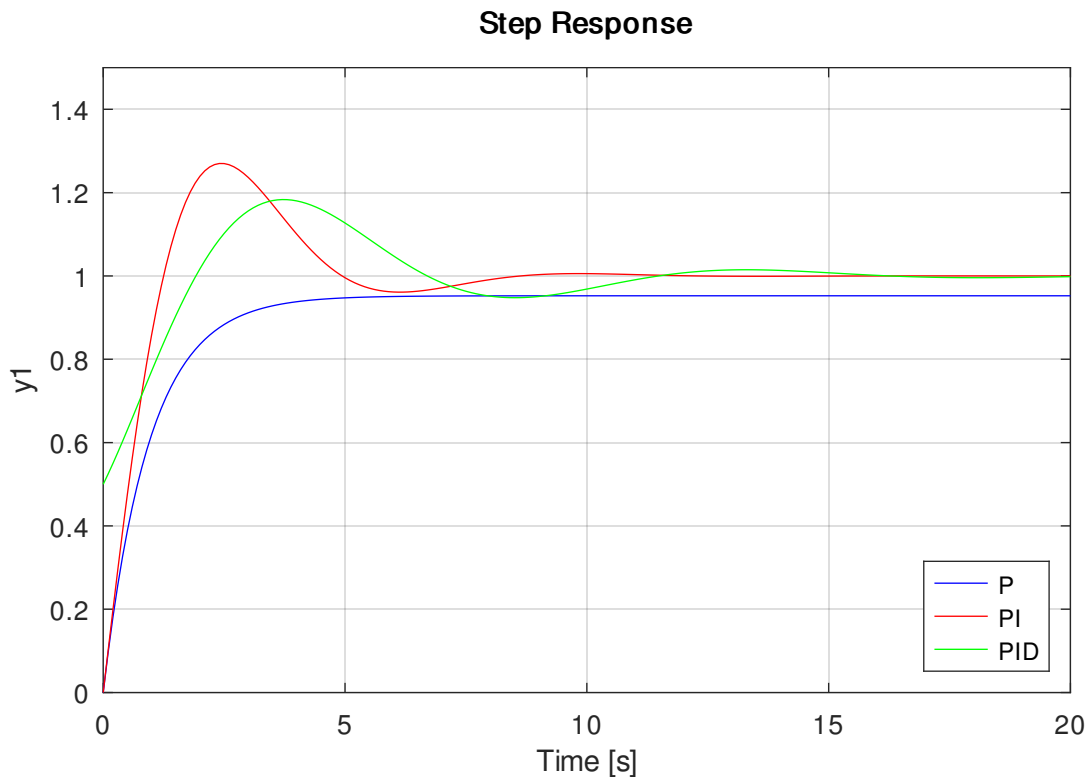


Figure 1: Step response of the cruise control system under P, PI, and PID control.

## Problem 2.A — Realistic Op Amp

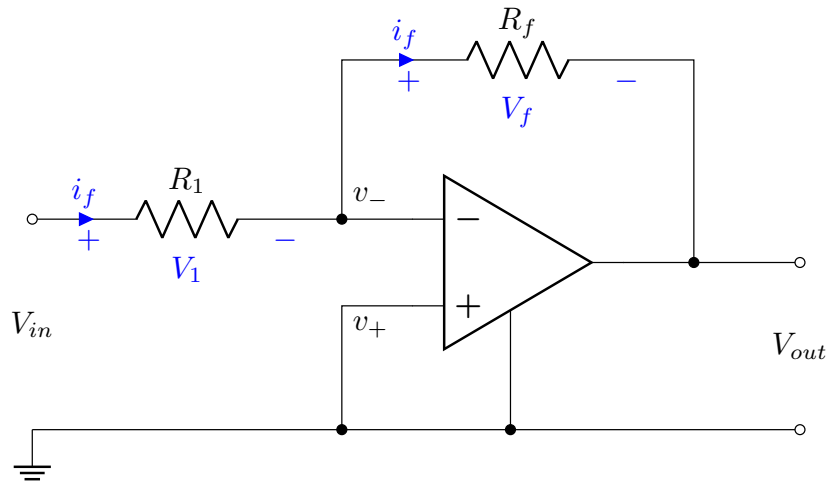


Figure 2: Inverting amplifier.

Start with the signals and parameters:

| Symbol    | Units | Description                     |
|-----------|-------|---------------------------------|
| $v_{in}$  | [V]   | Input voltage                   |
| $v_{out}$ | [V]   | Output voltage                  |
| $v_1$     | [V]   | Voltage across $R_1$            |
| $v_f$     | [V]   | Voltage across $R_f$            |
| $i_f$     | [A]   | Current through $R_1$ and $R_f$ |

Table 1: Symbols of Problem 2.A

| Parameter | Units                   | Description |
|-----------|-------------------------|-------------|
| $R_1$     | $[\Omega(\frac{V}{A})]$ | Resistance  |
| $R_f$     | $[\Omega(\frac{V}{A})]$ | Resistance  |

Table 2: Parameters of Problem 2.A

Symbols:

$$v_{in}(t) = v_{out}(t) + v_f(t) + v_1(t)$$

$$v_-(t) = v_{in} - v_1(t)$$

$$v_-(t) = v_{out}(t) + v_f(t)$$

$$v_1(t) = R_1 i_f(t)$$

$$v_f(t) = R_f i_f(t)$$

Introduce parameters:

$$v_{in} = v_{out} + R_f i_f + R_1 i_f$$

$$v_- = v_{in} - R_1 i_f$$

Transform:

$$V_{in}(s) = V_{out}(s) + (R_f + R_1)I_f(s)$$

$$V_-(s) = V_{in}(s) - R_1 I_f(s)$$

$$V_{out}(s) = -\frac{10^7}{s+1}V_-(s)$$

Substitute:

$$V_- = -10^{-7}(s+1)V_{out}$$

$$-10^{-7}(s+1)V_{out} = V_{in} - R_1 I_f$$

$$R_1 I_f = V_{in} + 10^{-7}(s+1)V_{out}$$

$$I_f = \frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + (R_f + R_1)\frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} - \left(\frac{R_f}{R_1} + 1\right)V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)10^{-7}(s+1)V_{out}$$

$$R_1 V_{in} - (R_f + R_1)V_{in} = R_1 V_{out} + (R_f + R_1)10^{-7}(s+1)V_{out}$$

$$-R_f V_{in} = (R_1 + 10^{-7}(R_f + R_1)(s+1))V_{out}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{R_1 + 10^{-7}(R_f + R_1)(s+1)}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{10^{-7}(R_f + R_1)s + 10^{-7}(R_f + R_1) + R_1}$$

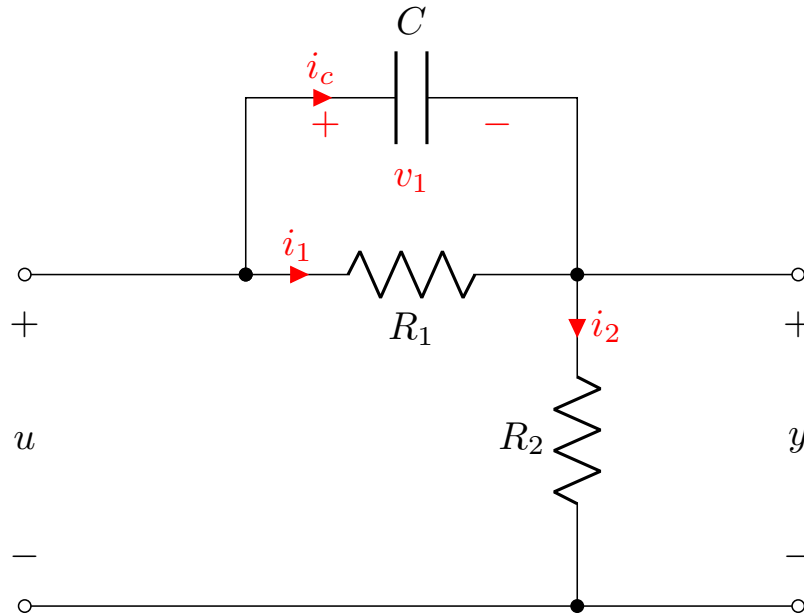
**Problem 2.B.a** — Passive lead circuit

Figure 3: Passive lead circuit.

| Symbol   | Units | Description                  |
|----------|-------|------------------------------|
| $u(t)$   | [V]   | Input voltage                |
| $y(t)$   | [V]   | Output voltage               |
| $v_1(t)$ | [V]   | Voltage across $R_1$ and $C$ |
| $i_1(t)$ | [A]   | Current through $R_1$        |
| $i_c(t)$ | [A]   | Current through $C$          |
| $i_2(t)$ | [A]   | Current through $R_2$        |

Table 3: Symbols of Problem 2.B.a

| Parameter | Units                        | Description |
|-----------|------------------------------|-------------|
| $R_1$     | $[\Omega(\frac{V}{A})]$      | Resistance  |
| $R_2$     | $[\Omega(\frac{V}{A})]$      | Resistance  |
| $C$       | $[F(\frac{A \cdot sec}{V})]$ | Capacitance |

Table 4: Parameters of Problem 2.B.a

Symbol equations:

$$u(t) = y(t) + v_1(t)$$

$$i_2(t) = i_1(t) + i_C(t)$$

Parameter equations:

$$i_c(t) = C\dot{v}_1(t)$$

$$i_2(t) = \frac{y(t)}{R_2}$$

$$i_1(t) = \frac{v_1(t)}{R_1}$$

Substitute:

$$\frac{y}{R_2} = \frac{v_1}{R_1} + C\dot{v}_1$$

$$R_1 y = R_2 v_1 + R_1 R_2 C \dot{v}_1$$

Frequency Domain:

$$U(s) = Y(s) + V_1(s)$$

$$R_1 Y(s) = (R_1 R_2 C s + R_2) V_1(s)$$

$$(R_1 R_2 C s + R_2) V_1(s) = R_1 Y(s)$$

$$V_1(s) = \frac{R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$U(s) = Y(s) + \frac{R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$U(s) = \left( \frac{R_1 R_2 C s + R_2}{R_1 R_2 C s + R_2} + \frac{R_1}{R_1 R_2 C s + R_2} \right) Y(s)$$

$$U(s) = \frac{R_1 R_2 C s + R_2 + R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$G_{yu}(s) = \frac{Y(s)}{U(s)} = \frac{R_1 R_2 C s + R_2}{R_1 R_2 C s + R_2 + R_1}$$

Units:

$$\frac{\frac{V}{A} \frac{V}{A} \frac{A \cdot \text{sec}}{V} \text{sec}^{-1} + \frac{V}{A} + \frac{V}{A}}{\frac{V}{A} \frac{V}{A} \frac{A \cdot \text{sec}}{V} \text{sec}^{-1} + \frac{V}{A}}$$